UM-2 field equations

The coupled field equations of Einstein-Lovelock gravity, the dirac16complex field, the gauge fields G, W^+, W^-, W^3 (Z, γ), B, the fermions ν, e, u, d and the tip sigma field, with every interaction

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Field equations, version 1, 2026-10-17

Repository once-ere/Dirac_claude, folder universe_model_2

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1 How to read this document

This document derives and states the complete set of coupled field equations of the universe model UM-2: the gravitational equations with every source, the equations of every matter field with every interaction, their boundary conditions, the exact background (order-0) equations, the exact reductions to the hidden coordinate x_0 that give the particle masses, the Kohn-Sham equations of dirac16complex coupled to all the other fields, and the linear (order ϵ) equations that describe the CMB, the CNB and the BAO. It is self-contained: every equation it uses is written here. Labels: AUTHOR (the author's notebooks), RECORD (the Revision record of this repository), STANDARD (textbook), ASSUMED, INVENTED (new in UM-2), DERIVED (computed and verified in UM-2; the verification is stated), OPEN.

1.1 Coordinates, signature, units

The author's coordinates are $X = \{x_0, x_1, \dots, x_7\}$: x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 three extra time-like directions. Frame metric $\eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ ($a = 0,\dots,7$ aligned with $x_0,\dots,x_7$); capital indices $M, N, \dots$ run over $x_0,\dots,x_7$, Greek μ,ν over the four observable coordinates $x_1,\dots,x_4$ (signature $(+,+,+,-)$). $z = 6Hx_0$; one period of x_0 is $\pi/(6H)$. Units $\hbar = c = 1$; mass dimensions in eight dimensions: $[\Psi] = [\psi] = 7/2$, $[\kappa] = -6$ ($\kappa = 8\pi G_8$).

1.2 Conventions

- The energy-momentum tensor is $T_{MN} = -\frac{2}{\sqrt{|g|}}\frac{\delta S_{matter}}{\delta g^{MN}}$ (vielbein variation for spinors), so that the energy density is $\rho = -T^{x_4}{}_{x_4}$ and the pressures are $p_M = T^M{}_M$ (no sum) for $M \neq x_4$; for a diagonal source $T^M{}_N = \mathrm{diag}(p_0, p_3, p_3, p_3, -\rho, p_t, p_t, p_t)$.
- Gauge fields are in the geometric normalisation: the couplings stand in front of the kinetic terms, $D_M = \partial_M + \Omega_M - iG^\alpha_M T^\alpha - iW^a_M \frac{\tau^a}{2} - iYB_M$, with $[G_M] = [W_M] = [B_M] = 1$ and $[\hat g^2] = -4$. Formulas that show $\hat g$, $\hat g'$, g, g' explicitly refer to the canonically normalised fields $W^a/\hat g$, $B/\hat g'$, $G^\alpha/\hat g_s$, written with the same letters. The weak bosons are $W^\pm = (W^1 \mp iW^2)/\sqrt{2}$: W^+ **carries the electric charge** $+e$ **and** W^- **the charge** $-e$; $Z = \cos\theta_W W^3 - \sin\theta_W B$, $A = \sin\theta_W W^3 + \cos\theta_W B$.
- The gamma matrices are the author's real 16×16 matrices of $Cl(4,4)$, $\{\gamma^a, \gamma^b\} = 2\eta^{ab}$; $\bar\Psi = \Psi^\dagger C$ with $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ (real, symmetric, $C^2 = 1$), $B = -iC\gamma^{x_4}$ (Hermitian, eigenvalues ±1, eight each), $\Gamma_4 = i\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$, $S^{ab} = \frac14[\gamma^a, \gamma^b]$. Since every $C\gamma^a$ is real and antisymmetric, $\bar\Psi\gamma^a\Psi$ is anti-Hermitian and the Hermitian vector currents are $i\bar\Psi\gamma^a\Psi$; $\bar\Psi\Psi$ is Hermitian.
- A delta function on the tip $y = -L$ in proper length is $\delta_{tip} = \delta(y + L)/f$.
- Notation: $\kappa = 8\pi G_8$ throughout; k denotes the block variable $1/f^2$, except in e^{ikx_1}, in h_j and in the dispersion $\varepsilon = e^{-a_4}|k|$, where it is the momentum of a mode.

2 The action

$$S = S_{EL} + S_{stab} + S_\Psi + S_{gauge} + S_f + S_{tip} + S_{I9} + S_{I11} + S_{I12}.$$

Gravity (AUTHOR weights):

$$S_{EL} = \frac{1}{2\kappa}\int d^8x\sqrt{|g|}\Big(\sum_{k=1}^{3}\alpha_k\mathcal{L}_{(k)} - 2\Lambda\Big), \qquad \alpha_k = w_k H^{2-2k}, \quad w = \Big(1, \frac{1}{32}, \frac{1}{1152}\Big), \quad \Lambda = -10H^2,$$

$$\mathcal{L}_{(k)} = \frac{1}{2^k}\,\delta^{A_1B_1\cdots A_kB_k}_{C_1D_1\cdots C_kD_k}\,R^{C_1D_1}{}_{A_1B_1}\cdots R^{C_kD_k}{}_{A_kB_k}: \qquad \mathcal{L}_{(1)} = R, \quad \mathcal{L}_{(2)} = R^2 - 4R_{AB}R^{AB} + R_{ABCD}R^{ABCD}.$$

Block stabiliser (ASSUMED; origin OPEN):

$$S_{stab} = -\int d^8x\sqrt{|g|}\,\rho_b(f), \qquad \kappa\rho_b(f) = \kappa\rho_b^*\Big(\frac{f_*}{f}\Big)^8, \qquad f = e^{-a_4}b_4^{1/2},$$

whose energy-momentum tensor is $T^M{}_N[\mathrm{stab}] = \mathrm{diag}(\rho_b, -\rho_b, -\rho_b, -\rho_b, -\rho_b, \rho_b, \rho_b, \rho_b)$.

dirac16complex (RECORD Lagrangian, canonical spin connection):

$$S_\Psi = \int d^8x\sqrt{|g|}\Big[\tfrac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m\bar\Psi\Psi - \frac{\lambda}{2}(\bar\Psi\Psi)^2\Big], \qquad m = 3HM = H,$$

$$D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \quad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M, \quad \Omega_M = \tfrac{1}{2}\omega_{Mab}S^{ab},$$
$$\omega_M{}^a{}_b = e^a{}_N\big(\partial_M e_b{}^N + \Gamma^N{}_{ML}e_b{}^L\big), \qquad \gamma^M = e_a{}^M\gamma^a.$$

Gauge fields (ASSUMED structure; $SU(3)_c \times SU(2)_L \times U(1)_Y$ in eight dimensions):

$$S_{gauge} = -\frac{1}{4}\int d^8x\sqrt{|g|}\Big(\frac{1}{\hat g_s^2}G^\alpha_{MN}G^{\alpha MN} + \frac{1}{\hat g^2}W^a_{MN}W^{aMN} + \frac{1}{\hat g'^2}B_{MN}B^{MN}\Big),$$

$$G^\alpha_{MN} = \partial_M G^\alpha_N - \partial_N G^\alpha_M + f^{\alpha\beta\gamma}G^\beta_M G^\gamma_N, \quad W^a_{MN} = \partial_M W^a_N - \partial_N W^a_M + \epsilon^{abc}W^b_M W^c_N, \quad B_{MN} = \partial_M B_N - \partial_N B_M.$$

Fermions (one generation; ASSUMED structure, boundary conditions INVENTED):

$$S_f = \sum_\psi \int d^8x\sqrt{|g|}\Big[\tfrac{1}{2}\big(\bar\psi\gamma^M D_M\psi - (D_M\bar\psi)\gamma^M\psi\big) - \mu_\psi\bar\psi\psi\Big], \qquad \psi \in \{Q, u_R, d_R, L, e_R, \nu_R\},$$

with $Q = (u,d)_L$ $(\mathbf{3},\mathbf{2},1/6)$, u_R $(\mathbf{3},\mathbf{1},2/3)$, d_R $(\mathbf{3},\mathbf{1},-1/3)$, $L = (\nu,e)_L$ $(\mathbf{1},\mathbf{2},-1/2)$, e_R $(\mathbf{1},\mathbf{1},-1)$, ν_R $(\mathbf{1},\mathbf{1},0)$; $Q = T_3 + Y$.

The tip sigma field (INVENTED; no Higgs particle), on the seven-dimensional tip $y = -L$:

$$S_{tip} = \int_{y=-L} d^7x\sqrt{|g_{tip}|}\Big[-\frac{v_\Sigma^2}{4}\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big) - \big(\bar L\,\Sigma\,\Lambda_\ell\,\ell_R + \bar Q\,\Sigma\,\Lambda_q\,q_R + \mathrm{h.c.}\big)\Big], \qquad \Sigma \in SU(2),$$

$$D_\mu\Sigma = \partial_\mu\Sigma - iW^a_\mu\frac{\tau^a}{2}\Sigma + iB_\mu\Sigma\frac{\tau^3}{2}, \qquad \ell_R = (\nu_R, e_R)^T, \quad q_R = (u_R, d_R)^T,$$
$$\Lambda_\ell = \mathrm{diag}(\lambda_\nu, \lambda_e), \qquad \Lambda_q = \mathrm{diag}(\lambda_u, \lambda_d),$$

invariant under $\Sigma \to U_L\Sigma U_Y^\dagger$, $U_Y = e^{i\theta\tau^3/2}$.

Portals (INVENTED):

$$S_{I9} = \int_{y=-L} d^7x\sqrt{|g_{tip}|}\,(c_Z + c'_Z\varsigma)\,(i\bar\Psi\gamma^\mu P_+\Psi)\,\mathcal{Z}_\mu, \qquad \mathcal{Z}_\mu = \frac{iv_\Sigma^2}{2}\mathrm{tr}\big(\Sigma^\dagger D_\mu\Sigma\,\tau^3\big), \qquad P_+ = \frac{1+\Gamma_4}{2}, \quad \varsigma = \pm 1,$$

$$S_{I11} = \int d^8x\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}\,(i\bar\Psi\gamma^M\Psi)\,\mathcal{J}_M, \qquad \mathcal{J}_M = \sum_\psi q_\psi\,(i\bar\psi\gamma_M\psi),$$

$$S_{I12} = -\int d^8x\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\,\mathcal{F}, \qquad \mathcal{F} = c_G G^\alpha_{MN}G^{\alpha MN} + c_W W^a_{MN}W^{aMN} + c_B B_{MN}B^{MN}.$$

In the unitary gauge $\Sigma = 1$: $\mathcal{Z}_\mu = \frac{v_\Sigma^2}{2}\sqrt{\hat g^2 + \hat g'^2}\,Z_\mu$, and the sigma-field kinetic term becomes $-\frac{v_\Sigma^2}{8}\big[2\hat g^2 W^+_\mu W^{-\mu} + (\hat g^2 + \hat g'^2)Z_\mu Z^\mu\big]$ (canonical fields).

3 The gravitational field equations

Varying S with respect to the metric gives

$$\sum_{k=1}^{3}\alpha_k\,E_{(k)}{}^M{}_N + \Lambda\,\delta^M_N = \kappa\,T^M{}_N, \qquad E_{(k)}{}^M{}_N = -\frac{1}{2^{k+1}}\,\delta^{MA_1B_1\cdots A_kB_k}_{NC_1D_1\cdots C_kD_k}\,R^{C_1D_1}{}_{A_1B_1}\cdots R^{C_kD_k}{}_{A_kB_k},$$

$E_{(1)}$ being the Einstein tensor. Every $E_{(k)}$ is divergence-free identically (DERIVED for the order-0 metric for $k = 1, 2, 3$), so the field equations imply $\nabla_M T^M{}_N = 0$, which holds on the matter field equations. The total source is

$$T^M{}_N = T^M{}_N[\mathrm{stab}] + T^M{}_N[\Psi] + T^M{}_N[\mathrm{gauge}] + T^M{}_N[f] + T^M{}_N[\mathrm{tip}] + T^M{}_N[I9] + T^M{}_N[I11] + T^M{}_N[I12].$$

3.1 The energy-momentum tensors

Stabiliser. ρ_b depends on the metric through the proper block volume $V_b \propto f^4$, $\rho_b \propto V_b^{-2}$; the variation gives $T_{\mu\nu} = -g_{\mu\nu}\rho_b$ for $\mu,\nu \in \{x_1,\dots,x_4\}$ and $T_{AB} = +g_{AB}\rho_b$ for $A,B \in \{x_0,x_5,x_6,x_7\}$, i.e. $\rho = \rho_b$, $p_3 = -\rho_b$, $p_0 = p_t = \rho_b$.

dirac16complex (and every Standard-Model fermion with $m \to \mu_\psi$, $\lambda \to 0$, D_M with gauge fields):

$$T_{MN}[\Psi] = -\frac{1}{2}\Big[\bar\Psi\gamma_{(M}D_{N)}\Psi - \big(D_{(M}\bar\Psi\big)\gamma_{N)}\Psi\Big] + g_{MN}\,\hat{\mathcal{L}}_\Psi, \qquad \hat{\mathcal{L}}_\Psi = \tfrac{1}{2}\big(\bar\Psi\gamma^P D_P\Psi - (D_P\bar\Psi)\gamma^P\Psi\big) - mS - \frac{\lambda}{2}S^2,$$

with $X_{(MN)} = \frac{1}{2}(X_{MN}+X_{NM})$ and $S = \bar\Psi\Psi$. On shell for a homogeneous state this gives $\rho = mS+U$, $p = SU'-U$ with $U = \frac{\lambda}{2}S^2$ (the record's convention, $\rho = -T^{x_4}{}_{x_4}$).

Gauge fields:

$$\begin{aligned}T_{MN}[\text{gauge}] =& \frac{1}{\hat g_s^2}\Big(G^\alpha_{MP}G^\alpha{}_N{}^P - \frac{1}{4}g_{MN}G^\alpha_{PQ}G^{\alpha PQ}\Big) + \frac{1}{\hat g^2}\Big(W^a_{MP}W^a{}_N{}^P - \frac{1}{4}g_{MN}W^a_{PQ}W^{aPQ}\Big)\\ &+ \frac{1}{\hat g'^2}\Big(B_{MP}B_N{}^P - \frac{1}{4}g_{MN}B_{PQ}B^{PQ}\Big).\end{aligned}$$

Tip (for μ,ν tangent to the tip; zero for the y components):

$$T_{\mu\nu}[\text{tip}] = \delta_{tip}\Big[\frac{v_\Sigma^2}{2}\,\mathrm{tr}\big((D_{(\mu}\Sigma)^\dagger D_{\nu)}\Sigma\big) + g_{\mu\nu}\,\hat{\mathcal{L}}_{tip}\Big], \qquad \hat{\mathcal{L}}_{tip} = -\frac{v_\Sigma^2}{4}\mathrm{tr}\big((D\Sigma)^\dagger D\Sigma\big) - \big(\bar L\Sigma\Lambda_\ell\ell_R + \bar Q\Sigma\Lambda_q q_R + \text{h.c.}\big).$$

Portals. The frame currents $i\bar\Psi\gamma^a\Psi$, $i\bar\psi\gamma^a\psi$ and the scalars $\bar\Psi\Psi$ do not depend on the vielbein, so only $\sqrt{|g|}$ and the explicit metric factors vary:

$$T_{MN}[I11] = g_{MN}\,\frac{c_V}{\Lambda_\star^6}(i\bar\Psi\gamma^P\Psi)\mathcal{J}_P,$$

$$T_{MN}[I12] = \frac{4\bar\Psi\Psi}{\Lambda_\star^3}\Big[c_G\Big(G^\alpha_{MP}G^\alpha{}_N{}^P - \frac{1}{4}g_{MN}G^\alpha_{PQ}G^{\alpha PQ}\Big) + (W,c_W) + (B,c_B)\Big],$$

so I12 acts as a field-dependent gauge coupling, $1/\hat g_i^2 \to 1/\hat g_i^2 + 4c_i\bar\Psi\Psi/\Lambda_\star^3$. The tip portal I9 contains the vielbein through $\gamma^\rho = e_a{}^\rho\gamma^a$ only ($\mathcal{Z}_\rho$ carries a lower index and no metric), so

$$T_{\mu\nu}[I9] = \delta_{tip}\,(c_Z + c'_Z\varsigma)\Big[g_{\mu\nu}\,(i\bar\Psi\gamma^\rho P_+\Psi)\mathcal{Z}_\rho - (i\bar\Psi\gamma_{(\mu}P_+\Psi)\,\mathcal{Z}_{\nu)}\Big].$$

4 The matter field equations

4.1 dirac16complex

$$\gamma^M D_M\Psi = \Big[m + \lambda\bar\Psi\Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big]\Psi - i\,\frac{c_V}{\Lambda_\star^6}\,\mathcal{J}_M\gamma^M\Psi - i\,\delta_{tip}\,(c_Z + c'_Z\varsigma)\,\mathcal{Z}_\mu\gamma^\mu P_+\Psi,$$

and its conjugate. The gravitational term is $\gamma^M\Omega_M$; for the order-0 metric (DERIVED, exact for every a_4, b_4 and on both halves of a period)

$$\gamma^M\Omega_M = \frac{3}{2}\,\frac{H}{f}\,\gamma^{x_0} + \Big(\frac{\dot b_4}{b_4} - \frac{\dot a_4}{2}\Big)\gamma^{x_4},$$

which vanishes only in flat 4+4 space ($H = 0$, constant a_4, b_4): the coupling of Ψ to gravity is non-trivial. Ψ is a singlet of the Standard-Model group: it couples to the Standard Model only through I9, I11, I12.

4.2 The Standard-Model fermions

For a multiplet ψ of charge q_ψ (I11),

$$\gamma^M D_M\psi = \mu_\psi\,\psi - i\,\frac{c_V}{\Lambda_\star^6}\,q_\psi\,(i\bar\Psi\gamma_M\Psi)\,\gamma^M\psi + \delta_{tip}\,(\text{Yukawa}),$$

$$(\text{Yukawa})_L = \Sigma\Lambda_\ell\,\ell_R, \quad (\text{Yukawa})_{\ell_R} = \Lambda_\ell\Sigma^\dagger L, \quad (\text{Yukawa})_Q = \Sigma\Lambda_q\,q_R, \quad (\text{Yukawa})_{q_R} = \Lambda_q\Sigma^\dagger Q.$$

The tip terms are equivalent to the self-adjoint mixing conditions at $y=-L$ (DERIVED; block form $\chi=(a,ib)$)

$$b_L(-L) = \kappa_f\, b_R(-L), \qquad a_R(-L) = -\kappa_f\, a_L(-L),$$

with κ_f the tip mixing of the fermion ($\kappa_f=1$ in UM-2, ASSUMED).

4.3 The gauge fields

With the fermion currents $j_s^{\alpha N} = \sum_q i\bar q\gamma^N T^\alpha q$, $j^{aN} = \sum_{\text{doublets}} i\bar\psi\gamma^N\frac{\tau^a}{2}\psi$, $j_Y^N = \sum_\psi Y_\psi\, i\bar\psi\gamma^N\psi$, the effective inverse couplings $\frac{1}{\hat g^2_{i,eff}} = \frac{1}{\hat g_i^2} + \frac{4c_i\bar\Psi\Psi}{\Lambda_\star^3}$ (I12), and in the unitary gauge $\Sigma=1$ (geometric fields),

$$D_M\Big(\frac{1}{\hat g^2_{s,eff}}G^{\alpha MN}\Big) = j_s^{\alpha N},$$

$$D_M\Big(\frac{1}{\hat g^2_{eff}}W^{1,2\,MN}\Big) = j^{1,2\,N} + \delta_{tip}\frac{v_\Sigma^2}{4}W^{1,2\,N},$$

$$D_M\Big(\frac{1}{\hat g^2_{eff}}W^{3\,MN}\Big) = j^{3N} + \delta_{tip}\Big[\frac{v_\Sigma^2}{4}(W^3-B)^N - (c_Z+c'_Z\varsigma)\frac{v_\Sigma^2}{2}(i\bar\Psi\gamma^N P_+\Psi)\Big],$$

$$\nabla_M\Big(\frac{1}{\hat g'^2_{eff}}B^{MN}\Big) = j_Y^N - \delta_{tip}\Big[\frac{v_\Sigma^2}{4}(W^3-B)^N - (c_Z+c'_Z\varsigma)\frac{v_\Sigma^2}{2}(i\bar\Psi\gamma^N P_+\Psi)\Big],$$

where N runs over the tip directions in the δ_{tip} terms. The tip terms are mass terms for $W^\pm$ and Z only: the combination $A=\sin\theta_W W^3+\cos\theta_W B$ (canonical fields) receives none, and the photon stays massless; the Ψ current enters only through Z. For an x_0 profile these tip terms become the Robin conditions of section 6.1.

4.4 The tip sigma field

Σ has no radial mode. Its equations of motion are the Noether identities of the gauge symmetry: they follow from the divergence of the tip parts of the gauge-field equations (STANDARD for a gauged non-linear sigma model), so in the unitary gauge $\Sigma=1$ they are not independent; its three modes are the longitudinal W^+, W^- and Z.

4.5 Boundary conditions

The variational principle with the fold reflection $R: z\to\pi-z$ and the regular tip at $y=-L$ gives:

- **gravity and gauge fields**: even (Neumann) or odd (Dirichlet) at the fold, $\partial_y u=0$ or $u=0$; at the tip the natural (Robin) condition from the tip terms;
- **dirac16complex**: fold $(1-\gamma^{x_0})\Psi=0$ (even) or $(1+\gamma^{x_0})\Psi=0$ (odd), both parities being modes of the full period; tip $(1-\gamma^{x_0})\Psi=0$;
- **Standard-Model fermions**, with $P_{sel}=\frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-i\gamma^{x_5}\gamma^{x_6}}{2}$: doublets ($\mu_\psi>0$) $P_{sel}(1-\gamma^{x_0})\psi=0$ and $(1-P_{sel})(1+\gamma^{x_0})\psi=0$ at the fold, $(1-\gamma^{x_0})\psi=0$ at the tip; singlets ($\mu_\psi<0$) the same with $\gamma^{x_0}\to-\gamma^{x_0}$; at the tip modified by the mixing conditions above.

Each condition is Lagrangian for the boundary form of the Dirac operator, so every operator is self-adjoint (DERIVED).

5 The background (order 0)

5.1 The metric, the variables and the curvature

The order-0 metric is the corrected primordial metric of the Revision record (version 2) in the author's coordinates,

$$\bar g = \mathrm{diag}\big(e^{-2a_4}b_4\cot^2 z,\ e^{2a_4},\ e^{2a_4},\ e^{2a_4},\ -1,\\ -e^{-2a_4}b_4(\sin^2 z)^{1/6},\ -e^{-2a_4}b_4(\sin^2 z)^{1/6},\ -e^{-2a_4}b_4(\sin^2 z)^{1/6}\big), \qquad z=6Hx_0,$$

with $a_4(x_4)$, $b_4(x_4) > 0$. With $y = \ln|\sin z|/(6H)$ ($dy = \cot z\, dx_0$) it reads

$$ds^2 = e^{2a_4} d\vec{x}^{\,2} - dx_4^2 + f^2\big[dy^2 - e^{2Hy} d\vec{x}_t^{\,2}\big], \qquad f = e^{-a_4} b_4^{1/2}, \quad k = \frac{1}{f^2}, \quad A = \dot{a}_4, \quad h = \frac{\dot{f}}{f} = \frac{\dot{b}_4}{2b_4} - \dot{a}_4,$$

dots denoting d/dx_4; $\dot{k} = -2kh$. $\sqrt{|\overline{g}|} = e^{3a_4} f^4 |\cot z| (\sin^2 z)^{1/4} = e^{3a_4} f^4 e^{3Hy} |dy/dx_0|$. The Riemann tensor is pair-diagonal ($R^{ab}{}_{cd} = 0$ unless $\{a,b\} = \{c,d\}$) and every sectional curvature $K_{ab} = R^{ab}{}_{ab}$ is one of (DERIVED, identical in both charts, independent of x_0)

$$K_{ij} = A^2, \quad K_{i4} = A^2 + \dot{A}, \quad K_{iB} = Ah, \quad K_{4B} = h^2 + \dot{h}, \quad K_{BB'} = h^2 - H^2 k,$$

$i, j \in \{1,2,3\}$, $B, B' \in \{0,5,6,7\}$. For a pair-diagonal Riemann tensor the Lovelock densities and tensors are (DERIVED; checked against the Ricci route, the explicit Gauss-Bonnet tensor, the Bianchi identities and the minisuperspace variation with a lapse)

$$\mathcal{L}_{(k)} = 2^k k! \sum_{k \text{ disjoint pairs}} \prod_{\text{pairs}} K_{ab}, \qquad E_{(k)}{}^{h}{}_{h} = -2^{k-1} k! \sum_{k \text{ disjoint pairs avoiding } h} \prod_{\text{pairs}} K_{ab}.$$

5.2 The Lovelock tensors of the background

Every off-diagonal component vanishes; the x_0 and the extra-time components coincide, $E_{(k)}{}^{x_0}{}_{x_0} = E_{(k)}{}^{x_5}{}_{x_5}$, and nothing depends on x_0 (DERIVED). The independent components are

$$E_{(1)}{}^{x_4}{}_{x_4} = 6kH^2 - 12Ah - 6h^2 - 3A^2$$

$$E_{(1)}{}^{x_1}{}_{x_1} = 6kH^2 - 8Ah - 4\dot{h} - 10h^2 - 2\dot{A} - 3A^2$$

$$E_{(1)}{}^{x_0}{}_{x_0} = 3kH^2 - 9Ah - 3\dot{h} - 6h^2 - 3\dot{A} - 6A^2$$

$$\begin{aligned} E_{(2)}{}^{x_4}{}_{x_4} =& 144AhkH^2 + 24kH^2h^2 + 72kA^2H^2 - 12H^4k^2 - 48hA^3 - 216A^2h^2 \\ & - 144Ah^3 - 12h^4 \end{aligned}$$

$$\begin{aligned} E_{(2)}{}^{x_1}{}_{x_1} =& 96AhkH^2 + 48\dot{h}kH^2 + 72kH^2h^2 + 48\dot{A}kH^2 + 72kA^2H^2 - 96Ah\dot{h} \\ & - 32A\dot{A}h - 48\dot{h}h^2 - 12H^4k^2 - 48\dot{A}h^2 - 32hA^3 - 16\dot{h}A^2 \\ & - 136A^2h^2 - 192Ah^3 - 60h^4 \end{aligned}$$

$$\begin{aligned} E_{(2)}{}^{x_0}{}_{x_0} =& 36AhkH^2 + 12\dot{h}kH^2 + 12kH^2h^2 + 36\dot{A}kH^2 + 72kA^2H^2 - 72Ah\dot{h} \\ & - 72A\dot{A}h - 12\dot{h}h^2 - 36\dot{A}h^2 - 108hA^3 - 36\dot{h}A^2 - 180A^2h^2 \\ & - 12\dot{A}A^2 - 108Ah^3 - 12h^4 - 12A^4 \end{aligned}$$

$$E_{(3)}{}^{x_4}{}_{x_4} = 864hkA^3H^2 + 1296kA^2H^2h^2 - 216A^2H^4k^2 - 1440A^3h^3 - 1080A^2h^4$$

$$\begin{aligned} E_{(3)}{}^{x_1}{}_{x_1} =& 576Ah\dot{h}kH^2 + 576A\dot{A}hkH^2 + 288\dot{A}kH^2h^2 + 576hkA^3H^2 + 288\dot{h}kA^2H^2 + 1008kA^2H^2h^2 \\ & + 576AkH^2h^3 - 144\dot{A}H^4k^2 - 864\dot{h}A^2h^2 - 216A^2H^4k^2 - 576A\dot{h}h^3 - 576A\dot{A}h^3 \\ & - 144\dot{A}h^4 - 576A^3h^3 - 1368A^2h^4 - 576Ah^5 \end{aligned}$$

$$\begin{aligned} E_{(3)}{}^{x_0}{}_{x_0} =& 432A\dot{A}hkH^2 + 648hkA^3H^2 + 216\dot{h}kA^2H^2 + 216kA^2H^2h^2 + 216\dot{A}kA^2H^2 + 216kA^4H^2 \\ & - 432h\dot{h}A^3 - 648\dot{h}A^2h^2 - 648\dot{A}A^2h^2 - 432A\dot{A}h^3 - 648A^4h^2 - 1224A^3h^3 \\ & - 648A^2h^4 \end{aligned}$$

5.3 The background field equations

With the author's weights and $\Lambda = -10H^2$ the three independent components are (DERIVED; the x_4 component is a constraint, the other two are linear in $\dot{A}$ and $\dot{h}$)

$$
\begin{aligned}
x_4: \quad & -\frac{15A^2h^4}{16H^4} - \frac{5A^3h^3}{4H^4} - \frac{3h^4}{8H^2} - \frac{9Ah^3}{2H^2} - \frac{27A^2h^2}{4H^2} + \frac{9kA^2h^2}{8H^2} \\
& - \frac{3hA^3}{2H^2} + \frac{3hkA^3}{4H^2} - 6h^2 + \frac{3kh^2}{4} - 12Ah + \frac{9Ahk}{2} \\
& - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 + 6kH^2 - \frac{3H^2k^2}{8} \\
& = -\kappa\rho
\end{aligned}
$$

$$
\begin{aligned}
x_1: \quad & \Big[-\frac{h^4}{8H^4} - \frac{Ah^3}{2H^4} - \frac{3h^2}{2H^2} + \frac{kh^2}{4H^2} - \frac{Ah}{H^2} + \frac{Ahk}{2H^2} \\
& \quad - 2 + \frac{3k}{2} - \frac{k^2}{8} \Big] \dot{A} \\
& + \Big[-\frac{Ah^3}{2H^4} - \frac{3A^2h^2}{4H^4} - \frac{3h^2}{2H^2} - \frac{3Ah}{H^2} + \frac{Ahk}{2H^2} - \frac{A^2}{2H^2} \\
& \quad + \frac{kA^2}{4H^2} - 4 + \frac{3k}{2} \Big] \dot{h} \\
& - \frac{Ah^5}{2H^4} - \frac{19A^2h^4}{16H^4} - \frac{A^3h^3}{2H^4} - \frac{15h^4}{8H^2} - \frac{6Ah^3}{H^2} + \frac{Akh^3}{2H^2} \\
& - \frac{17A^2h^2}{4H^2} + \frac{7kA^2h^2}{8H^2} - \frac{hA^3}{H^2} + \frac{hkA^3}{2H^2} - 10h^2 + \frac{9kh^2}{4} \\
& - 8Ah + 3Ahk - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 \\
& + 6kH^2 - \frac{3H^2k^2}{8} \\
& = \kappa p_3
\end{aligned}
$$

$$
\begin{aligned}
x_0: \quad & \Big[-\frac{3Ah^3}{8H^4} - \frac{9A^2h^2}{16H^4} - \frac{9h^2}{8H^2} - \frac{9Ah}{4H^2} + \frac{3Ahk}{8H^2} - \frac{3A^2}{8H^2} \\
& \quad + \frac{3kA^2}{16H^2} - 3 + \frac{9k}{8} \Big] \dot{A} \\
& + \Big[-\frac{9A^2h^2}{16H^4} - \frac{3hA^3}{8H^4} - \frac{3h^2}{8H^2} - \frac{9Ah}{4H^2} - \frac{9A^2}{8H^2} + \frac{3kA^2}{16H^2} \\
& \quad - 3 + \frac{3k}{8} \Big] \dot{h} \\
& - \frac{9A^2h^4}{16H^4} - \frac{17A^3h^3}{16H^4} - \frac{9A^4h^2}{16H^4} - \frac{3h^4}{8H^2} - \frac{27Ah^3}{8H^2} - \frac{45A^2h^2}{8H^2} \\
& + \frac{3kA^2h^2}{16H^2} - \frac{27hA^3}{8H^2} + \frac{9hkA^3}{16H^2} - \frac{3A^4}{8H^2} + \frac{3kA^4}{16H^2} - 6h^2 \\
& + \frac{3kh^2}{8} - 9Ah + \frac{9Ahk}{8} - 6A^2 + \frac{9kA^2}{4} - 10H^2 \\
& + 3kH^2 \\
& = \kappa\bar{p}
\end{aligned}
$$

Here $\bar{p} = (p_0 + 3p_t)/4$. Because $E^{x_0}{}_{x_0} = E^{x_5}{}_{x_5}$ identically, the order-0 metric admits only sources with $p_0 = p_t$ (and no off-diagonal components); for a source with $p_0 \neq p_t$ the variation of the action with respect to the common block scale f gives the equation above with $\bar{p}$, and $p_0 - p_t$ drives a block-shape mode outside the order-0 metric (OPEN). The source is conserved,

$$\dot{\rho} = -3A(\rho + p_3) - h(\rho + p_0) - 3h(\rho + p_t) = -3A(\rho + p_3) - 4h(\rho + \bar{p}).$$

The total source is the stabiliser ($\kappa\rho_b = \kappa\rho_b^*(k/k_*)^4$, $p_3 = -\rho_b$, $\bar{p} = \rho_b$) plus matter.

5.4 The static block and the stabiliser

For $h=0$, $\dot A=0$:

$$x_4:\ -3G(k)A^2+6H^2k(1-2w_2k)+\Lambda+\kappa\rho=0,\qquad x_1:\ \text{the same with }\ -\kappa p_3\ \text{for}\ \kappa\rho,$$

$$x_0:\ 6(12w_2k-1)A^2-\frac{12(w_2-18w_3k)}{H^2}A^4+3H^2k+\Lambda-\kappa\bar p=0,\qquad G(k)=1-24w_2k+72w_3k^2=\frac{k^2-12k+16}{16}.$$

Without sources and with $A=0$ the only static block is $k_0=1/(4w_2)=8$, $\Lambda_0=-3H^2/(4w_2)=-24H^2$, with $G(k_0)=-1$ (repulsive four-dimensional gravity). With the stabiliser and $A=0$:

$$\kappa\rho_b=10H^2-6H^2k+\tfrac{3}{8}H^2k^2=\kappa\bar p=3H^2k-10H^2\implies k_\pm=12\pm\tfrac{4}{3}\sqrt{51},$$

$$k_*=12+\tfrac{4}{3}\sqrt{51}=21.5219,\quad f_*=0.215556,\quad G(k_*)=\tfrac{20}{3}+\sqrt{51}=13.8081,\quad \kappa\rho_b^*=(26+4\sqrt{51})H^2.$$

5.5 The dynamical system, the radion and the response to matter

The x_1 and x_0 components are solved for $\dot A$ and $\dot h$; with $\dot k=-2kh$ and the constraint they form a closed system for (A,h,k). Linearised about $(0,0,k_*)$ (DERIVED):

$$\delta\dot A=J_{Ak}\delta k,\quad \delta\dot h=J_{hk}\delta k,\quad \delta\dot k=-2k_*\delta h,\qquad \omega_r^2=2k_*J_{hk}=\Big(\frac{31552}{2249}-\frac{2416\sqrt{51}}{6747}\Big)H^2=(3.38705H)^2,$$

eigenvalues $0,\pm i\omega_r$: the block is stable and oscillates (the radion), the zero mode being the direction δA (a small four-dimensional cosmological term). For the stabiliser family $\bar p_b=\nu\rho_b$, $\rho_b\propto k^{2(1+\nu)}$, the static block is stable for every $\nu\in\{1/4,1/2,3/4,1,5/4,3/2,2,3\}$ ($\omega_r=3.33H$ to $4.10H$). With matter sources to first order (A^2 counts as first order: the constraint gives $3G(k_*)A^2=\kappa\delta\rho$ because $\frac{d}{dk}[6H^2k(1-2w_2k)+\kappa\rho_b]$ vanishes at k_*):

$$\delta\dot A=J_{Ak}\,\delta k+s_{A3}\,\kappa\delta p_3+s_{Ap}\,\kappa\delta\bar p+q_A\,\kappa\delta\rho,$$

$$\delta\dot h=J_{hk}\,\delta k+s_{h3}\,\kappa\delta p_3+s_{hp}\,\kappa\delta\bar p+q_h\,\kappa\delta\rho,$$

$$\delta\ddot k+\omega_r^2\,\delta k=-2k_*\big(s_{h3}\,\kappa\delta p_3+s_{hp}\,\kappa\delta\bar p+q_h\,\kappa\delta\rho\big),$$

$$J_{Ak}=\tfrac{6(238-19\sqrt{51})}{2249}H^2,\quad J_{hk}=\tfrac{4063-485\sqrt{51}}{2249}H^2,\quad s_{A3}=\tfrac{2283}{4498}-\tfrac{162\sqrt{51}}{2249},\quad s_{Ap}=\tfrac{28\sqrt{51}-114}{2249},$$

$$q_A=\tfrac{135021}{265382}-\tfrac{10308\sqrt{51}}{132691},\quad s_{h3}=\tfrac{21\sqrt{51}}{2249}-\tfrac{171}{4498},\quad s_{hp}=\tfrac{239\sqrt{51}}{6747}-\tfrac{485}{2249},\quad q_h=\tfrac{57}{4498}-\tfrac{7\sqrt{51}}{2249}$$

(numerically $0.272956H^2$, $0.266522H^2$, -0.006853, 0.038221, -0.045997, 0.028666, 0.037320, -0.009555). In the adiabatic response ($\delta\ddot k=0$), $\delta k=(-0.107556\,\kappa\delta p_3-0.140028\,\kappa\delta\bar p+0.035852\,\kappa\delta\rho)/H^2$, and 3-space obeys exactly

$$\delta\dot A=-\frac{\kappa(\delta\rho+\delta p_3)}{2G(k_*)},$$

the four-dimensional Friedmann acceleration equation (DERIVED).

5.6 The four-dimensional effective gravity

On a product of a four-dimensional metric with the static block (constant curvature $K=-H^2k$) the Lovelock densities factorise, $\mathcal{L}_{(k)}=\sum_{j+l=k}\frac{k!}{j!\,l!}\mathcal{L}^{(4)}_{(j)}\mathcal{L}^{(int)}_{(l)}$, with $\mathcal{L}^{(int)}_{(l)}=(1,12K,24K^2,0)$, $\mathcal{L}^{(4)}_{(3)}\equiv0$ and $\mathcal{L}^{(4)}_{(2)}$ topological in four dimensions, so

$$\sum_k\alpha_k\mathcal{L}_{(k)}-2\Lambda-2\kappa\rho_b=G(k)\,R^{(4)}+(\text{topological})-2G(k)\Lambda_4,\qquad \Lambda_4=\frac{6H^2k(1-2w_2k)+\Lambda+\kappa\rho_b}{G(k)},$$

and the four-dimensional field equations are those of general relativity,

$$G^{(4)}_{\mu\nu}+\Lambda_4\,g^{(4)}_{\mu\nu}=8\pi G_N\,T^{(4)}_{\mu\nu},\qquad 8\pi G_N=\frac{\kappa}{G(k_*)V_\perp},$$

$$V_\perp=\frac{2f^4v_t(1-e^{-3\eta})}{3H},\qquad T^{(4)}_{\mu\nu}=\int dx_0\,dx_5\,dx_6\,dx_7\sqrt{|g_\perp|}\,T_{\mu\nu},$$

with $\Lambda_4=0$ at the static point (the observed dark energy is a tuned residual, $\Lambda_4/H^2\sim10^{-92}$, OPEN). The gauge couplings are $g_i^2=\hat g_i^2/V_\perp$.

6 The field equations in the hidden coordinate: the masses

On the static block every four-dimensional mass is an eigenvalue of the x_0 operator of its field (the author's requirement: the wave functions depend on x_0 periodically with period $\pi/(6H)$, through $y(z)$). With $t = y/L \in [-1, 0]$ on each half of a period, bosonic eigenvalues are $\Lambda = (m\,fL)^2$ and fermionic ones $e = \varepsilon\, fL$.

6.1 Gauge fields and the graviton

A four-dimensional component $A_\mu(x)u(y)$ (gauge $A_y = 0$) has the reduced action $\int_{-L}^{0} dy\, f^4 e^{3Hy}\big[u'^2/f^2 - m^2u^2\big]$ per half (times $1/\hat g^2$), hence (DERIVED)

$$-\frac{1}{f^2}e^{-3Hy}\partial_y\big(e^{3Hy}\partial_y u\big) = m^2 u \iff -e^{-3\eta t}\big(e^{3\eta t}u'\big)' = \Lambda u,$$

$$\text{fold: } u'(0) = 0 \text{ (even) or } u(0) = 0 \text{ (odd)}; \qquad \text{tip: } u'(-1) = \beta\, u(-1),$$

the tip condition being the natural boundary condition of the tip mass term: $\beta = 0$ for the photon and the gluons, β_W for W^+ and W^- (one value for both: they are particle and antiparticle), $\beta_Z = \beta_W/\cos^2\theta_W$ (on-shell angle, $\cos^2\theta_W = m_W^2/m_Z^2$) for the Z (the ratio of the coefficients $\hat g^2$ and $\hat g^2 + \hat g'^2$ of the tip mass terms). The transverse-traceless graviton obeys the same operator with $\beta = 0$ in Einstein gravity. With $u = e^{-3\eta t/2}w$, $q^2 = \Lambda - 9\eta^2/4$:

$$\text{even: } \Big[q^2 + \frac{3\eta}{2}\Big(\beta + \frac{3\eta}{2}\Big)\Big]\sin q - \beta q\cos q = 0, \qquad \text{odd: } q\cos q + \Big(\beta + \frac{3\eta}{2}\Big)\sin q = 0.$$

For $\beta = 0$: $\Lambda = 0$ (the flat profile; graviton, photon and gluons are exactly massless), $9\eta^2/4 + n^2\pi^2$ (even), $q\cot q = -3\eta/2$ (odd).

6.2 Fermions

With $\Psi = e^{ikx_1}e^{-\frac{3}{2}Hy}e^{a_4/2}b_4^{-1}V_\beta\,\chi(y, x_4)$ the eight-dimensional Dirac equation (canonical spin connection included) is EXACTLY (DERIVED, all blocks, general a_4, b_4, $M(y)$, $v(y)$)

$$\gamma^M D_M\Psi - \big(M(y) - iv(y)\gamma^{x_4}\big)\Psi = e^{ikx_1}e^{-\frac{3}{2}Hy}e^{a_4/2}b_4^{-1}\,\gamma^{x_4}V_\beta\big[\partial_4\chi + ih_j\chi\big],$$

$$h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + M(y)\sigma_2 + e^{-a_4}k\,\sigma_3\Big] + v(y),$$

so $i\partial_4\chi = h_j\chi$. In every block (j, s_2, s_3): $\gamma^{x_0} = \sigma_3$, $B = js_2$, $C = s_2\sigma_2$, $\Gamma_4 = -js_2\sigma_3$, $P_+ = (1 - B\sigma_3)/2$. For a Standard-Model multiplet $M = \mu_\psi$, $v = 0$; with $c = f\mu_\psi L$ the zero modes are $\chi = (e^{ct}, 0)$ (doublet, $\Gamma_4 = +1$) and $\chi = (0, e^{-ct})$ (singlet, $\Gamma_4 = -1$), with $h_j(e^{ct}, 0) = je^{-a_4}k\,(e^{ct}, 0)$ exactly (massless four-dimensional particles). The tip mixing conditions give the exact mass condition

$$D(e) = e^2 s_L s_R - \kappa_f^2\,(C_L - c_L s_L)(C_R + c_R s_R) = 0, \qquad s = \frac{\sinh w}{w}, \quad C = \cosh w, \quad w = \sqrt{c^2 - e^2},$$

$$e \simeq \kappa_f\,\zeta(c_L)\,\zeta(-c_R), \qquad \zeta(c)^2 = \frac{c\,e^{-c}}{\sinh c} \qquad (c_L > 0,\ c_R < 0).$$

7 dirac16complex: the Kohn-Sham equations coupled to all fields

The interacting many-body state of Ψ on the block is described by Kohn-Sham orbitals of

$$h_j\chi = \varepsilon\chi, \qquad M_{eff}(y) = m + \frac{15}{16}\lambda S(y) + \frac{\langle\mathcal{F}\rangle}{\Lambda_\star^3}, \qquad v(y) = -\frac{\lambda}{16}n(y) + \frac{c_V}{\Lambda_\star^6}\sum_\psi q_\psi n_\psi(y) + v_{I9}(y),$$

where $\langle\mathcal{F}\rangle = c_G\langle G^\alpha_{MN}G^{\alpha MN}\rangle + \ldots$ is the gauge-field condensate (in the QCD vacuum the gluon condensate, a uniform mass shift), n_ψ the number densities of the Standard-Model fermions ($\langle\bar\psi\gamma^{x_4}\psi\rangle = i\,n_\psi$; I11), and

$v_{I9}=\delta_{tip}(c_Z+c'_Z\varsigma)\mathcal{Z}_4$ acts at the tip on the P_+ component, i.e. on the upper component of the Ψ_A blocks ($B=-1$) only (I9). The densities are

$$n=\sum_o w\,g\,f_o\,n_o,\quad S=\sum_o w\,g\,f_o\,s_o,\qquad n_o=P\chi_o^\dagger\chi_o,\quad s_o=P\chi_o^\dagger j\sigma_2\chi_o,\quad t_o=P\chi_o^\dagger j\sigma_3\chi_o,\quad P=\frac{e^{-3Hy}}{\mathrm{Vol}_7},$$

with $w=1/2$ for each fold parity, $g=4r_3(n^2)$ the degeneracy of the momentum shell $\vec{k}=\Delta k\,\vec{n}$ (r_3 the number of representations of n^2 as a sum of three squares), $\mathrm{Vol}_7=(2\pi/\Delta k)^3 v_t f^4$, and occupations f_o (aufbau, or Mermin $f_o=1/(e^{(\varepsilon_o-\mu)/T}+1)$). The energy and the energy-momentum tensor are

$$E_{KS}=\sum_o g\,f_o\,\varepsilon_o-2\,\mathrm{Vol}_7\int_{-L}^{0}e^{3Hy}e_{int}\,dy,\qquad e_{int}=\frac{15}{32}\lambda S^2-\frac{1}{32}\lambda n^2,$$

$$\rho=\sum wgf\varepsilon\,n_o-e_{int},\qquad p_3=\sum wgf\,\frac{e^{-a_4|k|}}{3}t_o+e_{int},\qquad p_t=e_{int},$$

$$p_0=\sum wgf\big[(\varepsilon-v)n_o-M_{eff}s_o-e^{-a_4|k|}t_o\big]+e_{int},$$

which satisfy $E_{KS}=2\mathrm{Vol}_7\int e^{3Hy}\rho\,dy$ and the y-conservation law $\big[e^{3Hy}p_0\big]_{-L}^{0}=3H\int_{-L}^{0}e^{3Hy}p_t\,dy$ (DERIVED for every self-consistent state). Boundary conditions: fold $\chi_2(0)=0$ (even) or $\chi_1(0)=0$ (odd), tip $\chi_2(-L)=0$ (from $(1\mp\gamma^{x_0})\Psi=0$). The Kohn-Sham source enters the background equations through its block averages ρ, p_3 and $\bar{p}=(p_0+3p_t)/4$.

8 The linear perturbations (order ϵ): CMB, CNB, BAO

The x_0-independent ($n=0$) perturbations obey the four-dimensional Einstein equations of section 5.6 linearised about a flat Friedmann background; the $n\geq 1$ modes (Kaluza-Klein gravitons and gauge bosons, mass ≥ 99 TeV) and the radion (48 TeV) are absent at cosmological scales. In conformal time τ ($dx_4=a\,d\tau$, $a=e^{a_4}$, $\mathcal{H}=a'/a$) and the Newtonian gauge $ds^2=a^2\big[-(1+2\psi)d\tau^2+(1-2\phi)d\vec{x}^{\,2}\big]$ (UM-2: $\psi=\epsilon\Phi$, $\phi=\epsilon\Psi_N$), for a Fourier mode k (STANDARD, Ma and Bertschinger):

$$k^2\phi+3\mathcal{H}(\phi'+\mathcal{H}\psi)=-4\pi G_N a^2\delta\rho,\qquad k^2(\phi'+\mathcal{H}\psi)=4\pi G_N a^2(\bar{\rho}+\bar{P})\theta,$$

$$\phi''+\mathcal{H}(\psi'+2\phi')+(2\mathcal{H}'+\mathcal{H}^2)\psi+\frac{k^2}{3}(\phi-\psi)=4\pi G_N a^2\delta P,\qquad k^2(\phi-\psi)=12\pi G_N a^2(\bar{\rho}+\bar{P})\sigma,$$

summed over the species of UM-2:

- **dark matter** (Ψ_B odd quanta, cold): $\delta_c'=-\theta_c+3\phi'$, $\theta_c'=-\mathcal{H}\theta_c+k^2\psi$;
- **baryons** (u, d confined in p, n; electrons): $\delta_b'=-\theta_b+3\phi'$, $\theta_b'=-\mathcal{H}\theta_b+c_s^2k^2\delta_b+\frac{4\bar{\rho}_\gamma}{3\bar{\rho}_b}an_e\sigma_T(\theta_\gamma-\theta_b)+k^2\psi$;
- **photons**: $\delta_\gamma'=-\frac{4}{3}\theta_\gamma+4\phi'$, $\theta_\gamma'=k^2(\frac{1}{4}\delta_\gamma-\sigma_\gamma)+k^2\psi+an_e\sigma_T(\theta_b-\theta_\gamma)$, $F_{\gamma 2}'=2\sigma_\gamma'=\frac{8}{15}\theta_\gamma-\frac{3}{5}kF_{\gamma 3}-\frac{9}{5}an_e\sigma_T\sigma_\gamma+\frac{1}{10}an_e\sigma_T(G_{\gamma 0}+G_{\gamma 2})$, $F_{\gamma l}'=\frac{k}{2l+1}[lF_{\gamma(l-1)}-(l+1)F_{\gamma(l+1)}]-an_e\sigma_T F_{\gamma l}$ ($l\geq 3$), $G_{\gamma l}'=\frac{k}{2l+1}[lG_{\gamma(l-1)}-(l+1)G_{\gamma(l+1)}]+an_e\sigma_T[-G_{\gamma l}+\frac{1}{2}(F_{\gamma 2}+G_{\gamma 0}+G_{\gamma 2})(\delta_{l0}+\frac{\delta_{l2}}{5})]$;
- **the two Ψ_A Weyl species** (massless, decoupled with the neutrinos): $\delta'=-\frac{4}{3}\theta+4\phi'$, $\theta'=k^2(\frac{1}{4}\delta-\sigma)+k^2\psi$, $F_2'=2\sigma'=\frac{8}{15}\theta-\frac{3}{5}kF_3$, $F_l'=\frac{k}{2l+1}[lF_{l-1}-(l+1)F_{l+1}]$ ($l\geq 3$);
- ν_e (mass 0.05 eV, comoving momentum q, $\varepsilon=\sqrt{q^2+a^2m_\nu^2}$, Fermi-Dirac $f_0(q)$): $\Psi_0'=-\frac{qk}{\varepsilon}\Psi_1-\phi'\frac{d\ln f_0}{d\ln q}$, $\Psi_1'=\frac{qk}{3\varepsilon}(\Psi_0-2\Psi_2)-\frac{\varepsilon k}{3q}\psi\frac{d\ln f_0}{d\ln q}$, $\Psi_l'=\frac{qk}{(2l+1)\varepsilon}[l\Psi_{l-1}-(l+1)\Psi_{l+1}]$ ($l\geq 2$);
- the Ψ_B fold band is never thermalised and does not contribute.

The background is $\mathcal{H}^2=\frac{8\pi G_N}{3}a^2\bar{\rho}_{tot}+\frac{\Lambda_4}{3}a^2$, adiabatic initial conditions with the power spectrum $\mathcal{P}(k)=A_s(k/k_0)^{n_s-1}$. The relativistic content is $N_{eff}=3.044$ (ν_e plus the two Ψ_A species at $T_\nu=(4/11)^{1/3}T_\gamma$).

9 The coupled system and how it is solved

The fields couple as follows: gravity to everything (the canonical spin connection for Ψ and the fermions, the metric in every kinetic term, the stabiliser); Ψ to gravity, to itself (λ), to the fermion currents (I11), to the gauge-field

invariants (I12) and, at the tip, to Z (I9); the fermions to the gauge fields, to Σ at the tip and to Ψ (I11); the gauge fields to the fermions, to Σ at the tip, to Ψ through I12 (a field-dependent coupling) and, for Z, through I9. UM-2 solves the coupled system in the following hierarchy, each step feeding the next:

1. the background: the static block $(A, h, k) = (0, 0, k_*)$ with the stabiliser, its stability and its first-order response to every block-averaged source; the primordial approach to it from the full nonlinear equations;
2. the x_0 eigenvalue problems on the static block: all masses (gauge bosons, graviton, fermions, Ψ);
3. the Kohn-Sham equations of Ψ with the portal mean fields of the other fields and the self-interaction; the Kohn-Sham energy-momentum tensor returned to step 1 (the adiabatic block response);
4. the four-dimensional effective theory and its linear perturbations (CMB, CNB, BAO).

10 What was verified

- the Lovelock tensors of the order-0 metric by four routes, the Bianchi identities ($k = 1, 2, 3$), the record's equations and the minisuperspace variation (sympy, exact);
- the static block, the stabiliser, the radion, the stabiliser family, the first-order response and the 4D Friedmann acceleration equation, the dimensional reduction (sympy, exact);
- the exact Dirac reduction in all eight blocks, the boundary conditions, the zero modes, the tip mixing condition, the gauge x_0 operator and the gauge invariance of the tip terms;
- the x_0 spectra (mpmath; finite elements to 5×10^{-10}; shooting to 3.5×10^{-10});
- the Kohn-Sham solver (record reference states to 10^{-13}; energy identity; conservation law; Richardson; Hellmann- Feynman for every portal mean field).